

# THE DRINFELD–KOHNO BRIDGE FOR CHIRAL ALGEBRAS: KZ MONODROMY, QUANTUM GROUP $R$ -MATRICES, AND THE ORDERED BAR COMPLEX

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**ABSTRACT.** The Knizhnik–Zamolodchikov connection on conformal blocks of the affine Kac–Moody algebra  $\widehat{\mathfrak{g}}_k$  has monodromy valued in the braid group. The quantum group  $U_q(\mathfrak{g})$  has a universal  $R$ -matrix that also furnishes braid representations. That these representations coincide is the Drinfeld–Kohno theorem. We prove that the ordered chiral bar complex  $B^{\text{ord}}(V^k(\mathfrak{g}))$  provides the bridge: its collision residue  $r(z) = k\Omega/z$  (trace-form convention; the level prefix  $k$  is essential and  $r|_{k=0} = 0$ ) is simultaneously the infinitesimal datum underlying the KZ connection and the classical limit of the quantum  $R$ -matrix. The bar differential encodes the chiral OPE, the deconcatenation coproduct encodes the  $\mathbf{E}_1$  composition, and the monodromy of the resulting Kohno connection is the  $R$ -matrix.

We organise the comparison into four stages. At DK-0, the KZ monodromy on evaluation modules equals the quantum  $R$ -matrix representation (the classical theorem). At DK-1, the spectral Drinfeld strictification converts the  $A_\infty$  bar-complex structure into a genuine associative structure with spectral parameter, controlled by a cohomological obstruction that vanishes for all simple Lie algebras by root-space one-dimensionality. At DK-2, the dg-shifted Yangian  $Y_h^{\text{dg}}(\mathfrak{g})$  is identified as the line-side Koszul dual of the affine algebra. At DK-3, the strictification extends to all simple types. We verify the entire bridge on the Heisenberg algebra (where the  $R$ -matrix is scalar) and on  $\mathfrak{sl}_2$  (where the  $4 \times 4$   $R$ -matrix reproduces colored Jones polynomials via bar-complex monodromy).

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*Date:* April 21, 2026.

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*Remark 0.1* (Convention for  $q$ : Kazhdan–Lusztig vs. Drinfeld). This paper uses two closely related but distinct deformation parameters, and we fix the convention once and for all.

• **Canonical (Kazhdan–Lusztig).** We set

$$q := \exp(2\pi i/(k + h^\vee)).$$

All  $R$ -matrices, braiding monodromies, quantum-group identifications ( $U_q(\mathfrak{g})$ ,  $\text{Rep}_q(\mathfrak{g})$ , Kazhdan–Lusztig equivalence) and the default statements of DK-0, DK-1, DK-2, DK-3 use this  $q$  throughout.

- **Drinfeld.** Where the half-monodromy is displayed directly (singlet/triplet eigenvalue tables, half-braiding computations), we write

$$q_D := \exp(\pi i / (k + h^\vee)), \quad q = q_D^2.$$

The Drinfeld variable  $q_D$  appears ONLY in displays explicitly marked with the subscript D.

**Bridge to `en.koszul.duality.tex`.** Vol I’s `en.koszul.duality` uses the Kazhdan–Lusztig convention throughout; earlier drafts of the present paper had used both. After this note, every new occurrence of  $q$  defaults to Kazhdan–Lusztig unless explicitly marked  $q_D$ .

## 1. INTRODUCTION: TWO INCARNATIONS OF THE SAME OBJECT

**1.1. The problem.** The representation theory of the affine Kac–Moody algebra  $\widehat{\mathfrak{g}}_k$  at generic level  $k \neq -h^\vee$  produces a flat connection on the configuration space  $\text{Conf}_n(\mathbb{C})$  of  $n$  distinct points in the complex plane: the *Knizhnik–Zamolodchikov connection*

$$\nabla_{\text{KZ}} = d - \frac{1}{k + h^\vee} \sum_{1 \leq i < j \leq n} \Omega_{ij} \frac{d(z_i - z_j)}{z_i - z_j}, \quad (1.1)$$

where  $\Omega = \sum_a I^a \otimes I^a$  is the Casimir tensor in an orthonormal basis for the Killing form [6]. The connection one-form is  $\Omega_{ij} dz_{ij}/z_{ij}$ , not  $\Omega_{ij} d \log z_{ij}$ : the insertion point differentials  $dz_{ij} = d(z_i - z_j)$  appear linearly, and the Arnold form  $\omega_{ij} = d \log(z_i - z_j)$  is a bar-construction coefficient, not the connection form.

Independently, the quantum group  $U_q(\mathfrak{g})$  at  $q = e^{\pi i / (k + h^\vee)}$  produces braid group representations from its universal  $R$ -matrix  $\mathcal{R} \in U_q(\mathfrak{g})^{\widehat{\otimes} 2}$ . On any collection of finite-dimensional  $\mathfrak{g}$ -modules  $V_1, \dots, V_n$ , the braided monoidal structure of  $\text{Rep}_q(\mathfrak{g})$  defines a representation  $\rho_n^{U_q} : B_n \rightarrow \text{GL}(V_1 \otimes \dots \otimes V_n)$ .

The Drinfeld–Kohno theorem [4, 2] states that these two braid representations coincide:

$$\rho_n^{\text{KZ}} \simeq \rho_n^{U_q(\mathfrak{g})}. \quad (1.2)$$

The proof passes through a third object: the Drinfeld associator  $\Phi \in U(\mathfrak{g})^{\widehat{\otimes} 3}[[\hbar]]$ , constructed from iterated integrals on  $\text{Conf}_3(\mathbb{C})$ , which converts infinitesimal braid relations (the classical Yang–Baxter equation) into finite braid relations (the quantum Yang–Baxter equation).

**1.2. The bridge.** The purpose of this paper is to exhibit a fourth construction that unifies the three: the *ordered chiral bar complex*  $B^{\text{ord}}(V^k(\mathfrak{g}))$ .

The bar complex  $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ , where  $\bar{\mathcal{A}} = \ker(\varepsilon)$  is the augmentation ideal, is an  $E_1$ -chiral coassociative coalgebra with differential encoding

the chiral OPE and deconcatenation coproduct encoding the topological composition. It carries a collision residue

$$r_{ij}(z) = \frac{k \Omega_{ij}}{z} \quad (1.3)$$

in the trace-form convention. This collision residue is the seed from which all four objects grow:

- (1) It defines the *Kohno connection* on ordered configurations, whose restriction to  $\widehat{\mathfrak{g}}_k$  is the KZ connection (1.1).
- (2) Its monodromy around the braid generator  $\gamma_{ij}$  is the *R-matrix*  $R_{ij} = \mathcal{P} \exp(\oint_{\gamma_{ij}} r_{ij}(z) dz) = e^{k \Omega_{ij}}$  (in the trace form; equivalently  $e^{\Omega_{ij}/(k+h^\vee)}$  in the KZ normalization).
- (3) The classical Yang–Baxter equation  $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$  is equivalent to the bar differential squaring to zero:  $d_B^2 = 0$ .
- (4) The Drinfeld associator  $\Phi$  is constructed from iterated integrals of  $d \log(z_i - z_j)$  weighted by the collision residues, and the spectral Drinfeld strictification converts the  $A_\infty$  structure of the bar complex into the associative Yangian structure.

The unifying principle (Drinfeld’s vision): all four constructions are projections of a single object, the Maurer–Cartan element  $\Theta_{\mathcal{A}}$  in the  $\mathbf{E}_1$ -chiral convolution algebra. The classical *r*-matrix is the degree-2 projection; the associator is the degree-3 projection; the quantum *R*-matrix is the exponentiated degree-2 projection. The bar complex holds all degrees simultaneously.

**1.3. The four stages.** We organize the comparison into a four-stage ladder, each stage adding algebraic depth.

| Stage | Content                                                                 | Status |
|-------|-------------------------------------------------------------------------|--------|
| DK-0  | KZ monodromy = quantum <i>R</i> -matrix on evaluation modules           | Proved |
| DK-1  | Spectral Drinfeld strictification ( $A_\infty \rightarrow$ associative) | Proved |
| DK-2  | $Y_h^{\text{dg}}(\mathfrak{g})$ as line-side Koszul dual                | Proved |
| DK-3  | Complete strictification for all simple types                           | Proved |

DK-0 is the classical Drinfeld–Kohno theorem. DK-1 replaces the analytic associator by an algebraic strictification controlled by a cohomological obstruction. DK-2 identifies the resulting strict algebra with the dg-shifted Yangian. DK-3 verifies that the obstruction vanishes for all simple Lie algebras by exploiting root-space one-dimensionality.

**1.4. The pure-braid Orlik–Solomon route to Koszulity.** The Shelton–Yuzvinsky theorem asserts that the Orlik–Solomon algebra  $\text{OS}(\mathcal{A})$  of any supersolvable arrangement  $\mathcal{A}$  is Koszul. The pure-braid arrangement  $\mathcal{A}_n^{\text{PB}}$  (hyperplanes  $z_i = z_j$ ) is supersolvable by induction on  $n$ , hence the pure-braid

Orlik–Solomon algebra  $\text{OS}(\mathfrak{t}_n)$  is Koszul. This is the purely combinatorial input on which the chiral Drinfeld–Kohno programme rests.

**Theorem 1.1** (Pure-braid Orlik–Solomon Koszulity route). *Let  $\mathfrak{g}$  be simple and  $V^k(\mathfrak{g})$  the universal affine vertex algebra at non-critical level. Then:*

- (i)  $H^\bullet(B^{\text{ord}}(V^k(\mathfrak{g}))) \cong \text{OS}(\mathfrak{t}_n) \otimes V^k(\mathfrak{g})^{\otimes n}$  as  $\Sigma_n$ -twisted graded objects (the ordered bar concentrates in weight 1 on cohomology; PBW collapse);
- (ii) the collision residue  $r(z) = k\Omega_{\text{tr}}/z$  (trace-form) solves CYBE in the Knizhnik–Drinfeld sense;
- (iii) the Cherednik monodromy representation of the pure braid group factors through  $H^\bullet(B^{\text{ord}}(V^k(\mathfrak{g})))$ .

*Consequence: the ordered-Koszul property of  $V^k(\mathfrak{g})$  is witnessed combinatorially by  $\text{OS}(\mathfrak{t}_n)$  Koszulity, without analytic input beyond  $d \log$  absorption.*

**Lemma 1.2** ( $R$ -twisted  $\Sigma_n$ -descent). *Let  $\mathcal{A}$  be an ordered-Koszul chiral algebra with  $R$ -matrix  $R(z)$ . The ordered bar complex  $B^{\text{ord}}(\mathcal{A})$  descends to the symmetric bar  $B^\Sigma(\mathcal{A})$  via the  $R$ -twisted  $\Sigma_n$ -coinvariant functor*

$$\text{av}^R: B^{\text{ord}}(\mathcal{A}) \longrightarrow B^\Sigma(\mathcal{A}), \quad a_1 \otimes \cdots \otimes a_n \mapsto \frac{1}{n!} \sum_{\sigma \in \Sigma_n} R_\sigma(z) \cdot \sigma(a_1 \otimes \cdots \otimes a_n),$$

where  $R_\sigma(z)$  is the spectral- $R$ -matrix cocycle associated to the permutation  $\sigma$ . The plain averaging map  $\text{av}$  (with  $R_\sigma = 1$ ) is the classical limit  $\hbar \rightarrow 0$ . The  $R$ -twist factor is necessary for the descent to produce a chain map at  $\hbar \neq 0$ ; setting  $R = 1$  collapses the quantum-group structure.

**Corollary 1.3** (Chiral Kac–Kazhdan formal smoothness). *Combined with the Francis–Gaitsgory bar-cobar  $(\infty, 2)$ -equivalence lifted to the properad level, the pure-braid Orlik–Solomon Koszulity route (Theorem 1.1) and the  $R$ -twisted  $\Sigma_n$ -descent (Lemma 1.2) imply that the chiral Kac–Kazhdan category is formally smooth at the properad level: every conilpotent-complete factorization deformation of a Koszul-locus chiral algebra extends to all orders.*

*Remark 1.4* (Four Yangian types must be kept distinct). The Drinfeld–Kohno bridge relates three of the four Yangian notions (Definition ?? in the companion paper on three-parameter  $\hbar$ ):

- DK-0 is a statement about  $Y_\hbar(\mathfrak{g})$  (classical).
- DK-1 and DK-2 produce  $Y_\hbar^{\text{dg}}(\mathfrak{g})$  (dg-shifted).
- The chiral-bar incarnation along this paper is  $Y_\hbar(\mathfrak{g})^{\text{ch}}$  (chiral;  $E_1$ -chiral on  $X$ ).
- $Y_\hbar(\mathfrak{g})^{\text{spec}}$  (spectral) is the transposed-coordinate presentation used in the toroidal and CoHA incarnations; not touched by DK-0 through DK-3.

Conflating any two of these four incarnations (classical on  $\mathbb{R}$ ; dg-shifted on the formal disk; chiral on the curve  $X$ ; spectral on  $\mathbb{A}_u^1$ ) is a type error: each carries its own derived centre and its own  $R$ -matrix datum.

We illustrate the full bridge on two benchmark families: the Heisenberg algebra  $\mathcal{H}_k$  (Section 7), where the collision residue is scalar, the Kohno connection is abelian, the  $R$ -matrix is  $e^{kh/z}$ , and the monodromy is diagonal; and the affine algebra  $\widehat{\mathfrak{sl}}_2$  at level  $k$  (Section 8), where the  $4 \times 4$   $R$ -matrix on the fundamental representation produces colored Jones polynomials from bar-complex monodromy.

**1.5. Conventions.** Throughout,  $\mathfrak{g}$  denotes a finite-dimensional simple Lie algebra over  $\mathbb{C}$ ,  $h^\vee$  its dual Coxeter number,  $k \in \mathbb{C} \setminus \{-h^\vee\}$  a generic level, and  $V^k(\mathfrak{g})$  the universal affine vertex algebra at level  $k$ . All gradings are cohomological ( $|d| = +1$ ), and the desuspension satisfies  $|s^{-1}v| = |v| - 1$ .

**Warning 1.5** (Two  $r$ -matrix conventions). Two conventions for the affine classical  $r$ -matrix coexist in the literature and in this programme:

- (i) *Trace-form*:  $r(z) = k\Omega/z$ . At  $k = 0$ , the  $r$ -matrix vanishes (abelian limit). At  $k = -h^\vee$ , the  $r$ -matrix is  $-h^\vee\Omega/z$  (finite; the critical-level pathology is in the Sugawara construction, not in  $r$  itself).
- (ii) *KZ normalization*:  $r(z) = \Omega/((k + h^\vee)z)$ . At  $k = 0$ , the  $r$ -matrix is  $\Omega/(h^\vee z) \neq 0$  for non-abelian  $\mathfrak{g}$  (the Lie bracket persists). At  $k = -h^\vee$ , the  $r$ -matrix diverges (Sugawara singularity).

The bridge identity is  $k\Omega_{\text{tr}} = \Omega/(k + h^\vee)$  at generic  $k$ , where the subscript distinguishes the two Casimir normalizations. In this paper we use the KZ normalization for the connection (matching the standard KZ equation) and the trace-form for the bar-complex collision residue (where the level prefix is essential for the  $k = 0$  vanishing check).

## 2. THE KZ CONNECTION FROM THE BAR COMPLEX

**2.1. The ordered bar complex.** Let  $\mathcal{A}$  be a chiral algebra on a smooth curve  $X$ . The ordered bar complex is the cofree tensor coalgebra on the desuspended augmentation ideal:

$$B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}), \quad (2.1)$$

where  $\bar{\mathcal{A}} = \ker(\varepsilon)$  is the augmentation ideal (not  $\mathcal{A}$  itself) and  $|s^{-1}v| = |v| - 1$  is the desuspension. The differential  $d_B$  encodes the chiral OPE residues at the boundary strata of the Fulton–MacPherson compactification  $\overline{\text{FM}}_n^{\text{ord}}(\mathbb{C})$ . The deconcatenation coproduct  $\Delta: B^{\text{ord}}(\mathcal{A}) \rightarrow B^{\text{ord}}(\mathcal{A}) \otimes B^{\text{ord}}(\mathcal{A})$  is the  $E_1$  coassociative structure.

The bar complex is an  $E_1$ -chiral coassociative coalgebra. It does not carry  $\text{SC}^{\text{ch}, \text{top}}$  structure; the Swiss-cheese structure emerges on the derived chiral center  $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ , computed using the bar complex as a resolution.

**2.2. The collision residue.** The collision of two bar inputs  $a_i, a_j$  at the boundary stratum  $z_i = z_j$  of  $\overline{\text{FM}}_n^{\text{ord}}(\mathbb{C})$  produces a residue. The  $d \log$  kernel  $\eta_{ij} = d \log(z_i - z_j)$  absorbs one power of the OPE pole: if the OPE of  $\mathcal{A}$  has pole of order  $p$ , the collision residue  $r(z)$  has pole of order  $p - 1$ .

For the affine Kac–Moody algebra  $V^k(\mathfrak{g})$ , the OPE of current operators is

$$J^a(z) J^b(w) \sim \frac{k \delta^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}.$$

The double pole contributes the collision residue

$$r_{ij}(z_i - z_j) = \frac{k \Omega_{ij}}{z_i - z_j}, \quad (2.2)$$

in the trace-form convention, where  $\Omega_{ij} = \sum_a I_i^a \otimes (I_a)_j$  is the Casimir acting in the  $(i, j)$  tensor slots. The simple pole contributes the Lie bracket term  $r_0 = f^{ab}_c J^c$ , which is the constant term in the Laurent expansion.

*Remark 2.1* (Level prefix verification). In the trace-form convention, setting  $k = 0$  gives  $r(z)|_{k=0} = 0$ : the collision residue vanishes in the abelian limit, as it must, since the Heisenberg algebra at  $k = 0$  is trivial. Setting  $k = -h^\vee$  gives  $r(z)|_{k=-h^\vee} = -h^\vee \Omega/z$ , which is finite; the critical-level singularity manifests in the Sugawara construction (the denominator  $k + h^\vee = 0$ ), not in the  $r$ -matrix.

**2.3. The Kohno connection.** The collision residues assemble into a flat connection on the trivial bundle over  $\text{Conf}_n^{\text{ord}}(\mathbb{C})$  with fibre  $\bar{\mathcal{A}}^{\otimes n}$ .

**Construction 2.2** (The Kohno connection from the bar complex). Define the connection

$$\nabla = d - \sum_{1 \leq i < j \leq n} r_{ij}(z_i - z_j) d(z_i - z_j). \quad (2.3)$$

The connection one-form is  $r_{ij}(z_{ij}) dz_{ij}$ , where  $dz_{ij} = d(z_i - z_j)$  is the insertion-point differential. This is *not*  $r_{ij} d \log z_{ij}$ : the Arnold form  $d \log(z_i - z_j)$  is a bar-construction coefficient (the  $d \log$  kernel that absorbs one OPE pole), not the connection form.

The connection is flat:  $\nabla^2 = 0$ . The curvature

$$\nabla^2 = \sum_{i < j < k} ([r_{ij}, r_{ik}] + [r_{ij}, r_{jk}] + [r_{ik}, r_{jk}]) dz_{ij} \wedge dz_{jk}$$

vanishes if and only if the collision residues satisfy the classical Yang–Baxter equation

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0, \quad (2.4)$$

and this identity follows from  $d_B^2 = 0$  on the ordered bar complex: the codimension-two strata of  $\overline{\text{FM}}_n^{\text{ord}}(\mathbb{C})$  impose precisely (2.4) as the three-term relation on triple collisions.

**Proposition 2.3** (KZ connection from the Kohno connection). *For the affine algebra  $V^k(\mathfrak{g})$ , the Kohno connection (2.3) with collision residue (2.2) is*

$$\nabla_{\text{KZ}} = d - \frac{1}{k + h^\vee} \sum_{1 \leq i < j \leq n} \Omega_{ij} \frac{d(z_i - z_j)}{z_i - z_j}, \quad (2.5)$$

where we have passed to the KZ normalization via  $k\Omega_{\text{tr}} = \Omega/(k + h^\vee)$ .

*Proof.* Substitute the trace-form collision residue (2.2) into the Kohno connection (2.3):

$$\nabla = d - \sum_{i < j} \frac{k\Omega_{ij}}{z_i - z_j} d(z_i - z_j) = d - \sum_{i < j} k\Omega_{ij} \frac{dz_{ij}}{z_{ij}}.$$

In the KZ normalization,  $k\Omega_{\text{tr}} = \Omega_{\text{KZ}}/(k + h^\vee)$ , giving (2.5). This is the Knizhnik–Zamolodchikov connection [6]. The  $d\tau$  term of the Knizhnik–Zamolodchikov–Bernard connection is absent: we are at genus 0.  $\square$

**2.4. The  $R$ -matrix as monodromy.** The monodromy of the Kohno connection produces the  $R$ -matrix.

**Theorem 2.4** (Bar-complex  $R$ -matrix). *At  $n = 2$ , the ordered configuration space  $\text{Conf}_2^{\text{ord}}(\mathbb{C}) = \{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \neq z_2\}$  has fundamental group  $\pi_1 = \mathbb{Z}$ , generated by the loop  $\gamma$  in which  $z_1$  circles  $z_2$ . The monodromy of the Kohno connection (2.3) around  $\gamma$  is*

$$R_{12} = \text{Mon}_\gamma(\nabla) = \mathcal{P} \exp\left(\oint_\gamma r_{12}(z) dz\right) = e^{\Omega_{12}/(k+h^\vee)}, \quad (2.6)$$

where  $\mathcal{P}$  denotes path-ordering. At general  $n$ , the pure braid group  $P_n = \pi_1(\text{Conf}_n^{\text{ord}}(\mathbb{C}))$  has standard generators  $\gamma_{ij}$  ( $i < j$ ), and the flat connection  $\nabla$  induces a monodromy representation

$$\rho: P_n \rightarrow \text{GL}(\bar{\mathcal{A}}^{\otimes n}), \quad \gamma_{ij} \mapsto R_{ij}.$$

The Yang–Baxter equation  $R_{12} R_{13} R_{23} = R_{23} R_{13} R_{12}$  is the exponentiated form of the CYBE (2.4), so  $\rho$  is a well-defined representation.

*Proof.* The monodromy of the flat connection  $\nabla = d - r_{12}(z) dz$  around  $\gamma$  (the loop  $z_1 - z_2 = \epsilon e^{2\pi it}$ ,  $t \in [0, 1]$ ) is the path-ordered exponential

$$R_{12} = \mathcal{P} \exp\left(\int_0^1 \frac{k\Omega_{12}}{\epsilon e^{2\pi it}} (2\pi i \epsilon e^{2\pi it}) dt\right) = \mathcal{P} \exp(2\pi i k \Omega_{12}).$$

Since  $\Omega_{12}$  acts as a single endomorphism (not a sum of  $z$ -dependent terms), the path-ordering is trivial and  $R_{12} = e^{2\pi i k \Omega_{12}}$  in the trace normalization where  $r_{\text{tr}}(z) = k\Omega/z$ . In the Kazhdan–Lusztig (KZ) normalization where  $r_{\text{KZ}}(z) = \Omega/((k + h^\vee)z)$ , the monodromy reads

$$R_{12} = \exp\left(\frac{2\pi i}{k + h^\vee} \Omega_{12}\right) = q^{\Omega_{12}}, \quad q := \exp\left(\frac{2\pi i}{k + h^\vee}\right).$$

The factor of  $2\pi i$  in the exponent is *structural*, not a contour normalization: it governs the braiding monodromy and identifies the Kazhdan–Lusztig root of unity  $q$  at integer level. In particular, at level  $k$  a positive integer,  $q$  is a primitive  $(k + h^\vee)$ -th root of unity and the braiding restricts to the quantum-group  $\text{Rep}_q(\mathfrak{g})$  on the evaluation-generated core (Kazhdan–Lusztig equivalence).



The Yang–Baxter equation follows from flatness:  $\nabla^2 = 0$  implies the CYBE (2.4), and the exponential map sends the CYBE to the YBE.  $\square$

*Remark 2.5* ( $R$ -twisted descent). The symmetric bar complex is the  $R$ -twisted  $\Sigma_n$ -descent of the ordered bar complex:  $B^\Sigma(\mathcal{A})_n \simeq (B^{\text{ord}}(\mathcal{A})_n)^{R\text{-}\Sigma_n}$ . The symmetric group acts on adjacent transpositions by  $\sigma_i \cdot (a_1 \otimes \cdots \otimes a_n) = (\tau_{i,i+1} \circ R_{i,i+1})(a_1 \otimes \cdots \otimes a_n)$ . This action is well-defined precisely when  $R$  satisfies the YBE; it factors through  $\Sigma_n$  precisely when  $R$  also satisfies strong unitarity  $R_{12}(z) R_{21}(-z) = \text{id}$ .

### 3. DK-0: KZ MONODROMY EQUALS QUANTUM $R$ -MATRIX

**3.1. Statement.** The classical Drinfeld–Kohno theorem identifies the KZ monodromy representation with the quantum group braid representation.

**Theorem 3.1** (Drinfeld–Kohno, genus 0). *Let  $\mathfrak{g}$  be a simple Lie algebra,  $k \in \mathbb{C} \setminus \{-h^\vee\}$  generic, and  $V_1, \dots, V_n$  finite-dimensional  $\mathfrak{g}$ -modules. On the evaluation-generated core  $\text{Core}_{\text{eval}}(\widehat{\mathfrak{g}}_k)$ , the monodromy representation*

$$\rho_n^{\text{KZ}} : \pi_1(\text{Conf}_n(\mathbb{C})) \longrightarrow \text{GL}(V_1 \otimes \cdots \otimes V_n)$$

*of the KZ connection (2.5) is isomorphic to the braid group representation obtained from the braided monoidal structure of  $\text{Rep}_q(\mathfrak{g})$  via the universal  $R$ -matrix of  $U_q(\mathfrak{g})$ :*

$$\rho_n^{\text{KZ}} \simeq \rho_n^{U_q(\mathfrak{g})}, \quad (3.1)$$

where  $q = e^{\pi i/(k+h^\vee)}$ .

**3.2. The Drinfeld associator.** The proof passes through three steps, each with a clear bar-complex interpretation.

**Construction 3.2** (The Drinfeld associator from iterated integrals). Drinfeld constructs a formal series  $\Phi \in U(\mathfrak{g})^{\otimes 3}[[\hbar]]$  satisfying the pentagon and hexagon equations, using iterated integrals of  $d \log(z_i - z_j)$  on  $\text{Conf}_3(\mathbb{C})$ . The Arnold forms  $\omega_{ij} = d \log(z_i - z_j)$  satisfy the Arnold relation  $\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0$ , and the iterated integrals along the real interval  $(0, 1)$  are the coefficients of  $\Phi$ .

In the bar-complex language, the Arnold forms are the propagators of the ordered bar differential, and the iterated integrals are the genus-0 ordered chiral homology amplitudes at degree 3. The pentagon equation for  $\Phi$  is the degree-3 component of the Maurer–Cartan equation  $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$  for the universal MC element  $\Theta_A$ .

*Proof of Theorem 3.1.* The proof proceeds in three steps.

*Step 1: Infinitesimal braid relations.* The collision residues  $r_{ij}(z) = k\Omega_{ij}/z$  satisfy the CYBE (2.4). This is the *infinitesimal braid relation*: the Lie algebra  $\mathfrak{t}_n$  generated by  $\{t_{ij}\}_{1 \leq i < j \leq n}$  with relations  $[t_{ij}, t_{kl}] = 0$  (for  $\{i, j\} \cap \{k, l\} = \emptyset$ ) and the infinitesimal YBE  $[t_{ij}, t_{ik} + t_{jk}] = 0$  is the Drinfeld–Kohno Lie algebra. The KZ connection defines a flat  $\mathfrak{t}_n$ -connection,

and its monodromy is a representation of  $\pi_1(\text{Conf}_n(\mathbb{C}))$  that factors through the pure braid group  $P_n$ .

*Step 2: The associator bridge.* The Drinfeld associator  $\Phi \in \exp(\hat{\mathfrak{f}}_2)$  (where  $\hat{\mathfrak{f}}_2$  is the completed free Lie algebra on two generators  $t_{12}, t_{23}$ ) satisfies the pentagon and hexagon equations. It provides an equivalence of braided monoidal categories between the completion  $\hat{\mathfrak{t}}_n\text{-mod}$  (with KZ braiding) and  $\text{Rep}_q(\mathfrak{g})$  (with  $R$ -matrix braiding) at  $q = e^{\hbar}$ ,  $\hbar = \pi i/(k + h^\vee)$ .

The pentagon equation for  $\Phi$  states that the two parenthesizations  $(V_1 \otimes V_2) \otimes V_3 \rightarrow V_1 \otimes (V_2 \otimes V_3)$  agree after composing the five natural maps around the Stasheff pentagon. This is precisely the degree-3 coherence condition on the MC element  $\Theta_{\mathcal{A}}$  in the  $E_1$ -chiral convolution algebra: the bar complex packages the pentagon into  $d_B^2 = 0$  at degree 3.

*Step 3: One-loop collapse.* For the affine algebra  $V^k(\mathfrak{g})$ , the higher  $A_\infty$  operations  $m_k$  ( $k \geq 3$ ) vanish on the evaluation-generated core  $\text{Core}_{\text{eval}}(\hat{\mathfrak{g}}_k)$ . (This is the one-loop exactness of the BV–BRST differential on the eval core: only the quadratic part of the chiral action contributes; see Vol I, Theorem ?? in `five_theorems_modular_koszul.tex`.) The full superconnection on the bar complex therefore collapses to the Kohn connection, and the bar-complex monodromy equals the KZ monodromy:  $\rho_n^{\text{bar}} = \rho_n^{\text{KZ}}$ . Combining with Step 2:  $\rho_n^{\text{KZ}} \simeq \rho_n^{U_q(\mathfrak{g})}$ .  $\square$

**3.3. The DK square.** The Drinfeld–Kohno theorem at the level of the bar complex fits into a commutative square that relates Koszul duality to the quantum group involution  $q \mapsto q^{-1}$ .

**Theorem 3.3** (The DK square). *For  $\mathfrak{g}$  simple and  $k$  generic, the following square commutes on the evaluation-generated core  $\text{Core}_{\text{eval}}(\hat{\mathfrak{g}}_k)$ :*

$$\begin{array}{ccc} D^b(\text{Rep}_{\text{fd}}^{\text{eval}}(Y_{\hbar}(\mathfrak{g}))) & \xrightarrow{\Phi} & D^b(\text{Rep}_{\text{fd}}^{\text{eval}}(Y_{-\hbar}(\mathfrak{g}))) \\ \Phi_K \downarrow & & \downarrow \Phi_K \\ D^b(\text{Rep}_{\text{fd}}(U_q(\mathfrak{g}))) & \xrightarrow{q \mapsto q^{-1}} & D^b(\text{Rep}_{\text{fd}}(U_{q^{-1}}(\mathfrak{g}))) \end{array} \quad (3.2)$$

where:

- (i)  $\Phi$  is the  $E_1$ -chiral Koszul duality functor (bar-cobar on the ordered configurations), which reverses the braiding:  $R(u; \hbar) \mapsto R^{-1}(u; \hbar) = R(u; -\hbar)$ .
- (ii)  $\Phi_K$  is the Kazhdan equivalence  $\text{Rep}_{\text{fd}}(Y_{\hbar}(\mathfrak{g})) \xrightarrow{\sim} \text{Rep}_{\text{fd}}(U_q(\mathfrak{g}))|_{q=e^{\hbar}}$ , the monoidal equivalence that sends the rational  $R$ -matrix of the Yangian to the trigonometric  $R$ -matrix of the quantum group.
- (iii) The bottom arrow is the Hopf algebra involution  $q \mapsto q^{-1}$ , sending  $\mathcal{R}_q \mapsto \mathcal{R}_q^{-1} = \mathcal{R}_{q^{-1}}$ .

*The commutativity states: Koszul duality on ordered configuration spaces geometrically implements the quantum group involution. The algebraic operation  $q \mapsto q^{-1}$  is realized by Verdier duality, which reverses the orientation of collision cycles.*

*Proof.* The top arrow sends evaluation modules  $V(a)_h \mapsto V(-a)_{-h}$  and reverses the braiding.

The left vertical (Kazhdan) sends  $V(a)_h$  to  $V_q(e^a)$ . The bottom arrow sends  $V_q(e^a) \mapsto V_{q^{-1}}(e^a)$ .

Following the right vertical first:  $V(-a)_{-h} \mapsto V_{q^{-1}}(e^{-a})$ . Upon restriction to  $U_{q^{-1}}(\mathfrak{g})$  (forgetting the loop algebra evaluation point),  $V_{q^{-1}}(e^{-a})$  and  $V_{q^{-1}}(e^a)$  carry the same highest-weight representation. The square commutes at the level of  $U_{q^{-1}}(\mathfrak{g})$ -module categories.

The braiding compatibility follows from the geometric content of the top arrow: Verdier duality on  $\overline{\text{FM}}_n^{\text{ord}}(\mathbb{C})$  reverses the orientation of the collision cycle  $\gamma_{12}$ , sending the KZ monodromy  $\sigma \mapsto \sigma^{-1}$ . Under the Kazhdan identification, this is  $\mathcal{R}_q \mapsto \mathcal{R}_q^{-1} = \mathcal{R}_{q^{-1}}$ .  $\square$

**3.4. Four-path verification for  $\mathfrak{sl}_2$ .** The DK-0 identification can be verified from four independent constructions, all producing the same braid group representation on  $V_{j_1} \otimes \cdots \otimes V_{j_n}$  for  $\widehat{\mathfrak{sl}}_2$  at level  $k$ :

- (1) KZ monodromy of  $\nabla_{\text{KZ}}$ ;
- (2) quantum Casimir eigenvalues  $q^{C_2}$  at  $q = e^{\pi i/(k+2)}$ ;
- (3) Yangian  $R$ -matrix  $R^Y(u) = u \cdot I + iP$  at the Drinfeld specialization  $u_D = \cot(\pi/(k+2))$ ;
- (4) Verlinde fusion truncation at the Weyl alcove boundary.

For  $n = 4$  with all external legs in the fundamental representation  $V_{1/2}$ , the KZ equation reduces to the hypergeometric ODE

$$z(1-z)F'' + (c_h - (a+b+1)z)F' - abF = 0 \quad (3.3)$$

with  $a = -1/(k+2)$ ,  $b = (k+1)/(k+2)$ ,  $c_h = k/(k+2)$ . The two independent solutions

$$F_0(z) = {}_2F_1\left(-\frac{1}{k+2}, \frac{k+1}{k+2}; \frac{k}{k+2}; z\right) \quad (\text{singlet channel}), \quad (3.4)$$

$$F_1(z) = z^{2/(k+2)} {}_2F_1\left(\frac{1}{k+2}, \frac{k+3}{k+2}, \frac{k+4}{k+2}; z\right) \quad (\text{triplet channel}), \quad (3.5)$$

are the  $\mathfrak{sl}_2$  conformal blocks in the  $s$ -channel. Their braiding eigenvalues in the singlet and triplet channels are  $R_0 = -q^{-3/2}$  and  $R_1 = q^{1/2}$ , where  $q = e^{i\pi/(k+2)}$ .

#### 4. DK-1: SPECTRAL DRINFELD STRICTIFICATION

**4.1. The strictification problem.** The ordered bar complex of the affine algebra  $V^k(\mathfrak{g})$  carries an  $A_\infty$  structure: the sequence of operations  $\{m_k\}_{k \geq 1}$  on  $B^{\text{ord}}(V^k(\mathfrak{g}))$  encodes the multi-point OPE residues at all collision orders. At the level of monodromy (DK-0), the higher operations  $m_k$  for  $k \geq 3$  do not contribute on evaluation modules. At the chain level, they are present

and obstruct the passage from  $A_\infty$  to a strict associative structure with spectral parameter.

The spectral Drinfeld strictification converts this  $A_\infty$  structure into a genuine associative product. The obstruction is cohomological: at each filtration level, a class in a spectral cochain complex either vanishes (allowing strictification to proceed) or survives (blocking the comparison). For all simple Lie algebras, the obstruction vanishes.

#### 4.2. The filtered $A_\infty$ structure.

**Definition 4.1** (Filtered bar-complex  $A_\infty$  structure). The ordered bar complex  $B^{\text{ord}}(V^k(\mathfrak{g}))$  carries a filtration  $F^\bullet$  by collision depth:  $F^p B^{\text{ord}}$  consists of elements supported on configurations where at most  $p$  points collide. The  $A_\infty$  operations respect this filtration:  $m_k: (F^{p_1} B) \otimes \cdots \otimes (F^{p_k} B) \rightarrow F^{p_1 + \cdots + p_k + k - 1} B$ .

The associated graded  $\text{gr}^p B^{\text{ord}}$  inherits a differential from  $m_1$  and higher operations from the  $m_k$ . At the bottom filtration ( $p = 1$ ), the structure is strict:  $\text{gr}^1 B^{\text{ord}} = s^{-1} \bar{\mathcal{A}}$  with the identity product. The first non-trivial  $A_\infty$  data appears at  $p = 2$ .

**4.3. The spectral cochain complex.** The obstruction to strictification is controlled by a spectral cochain complex that measures the failure of the  $A_\infty$  quasi-associator to be trivializable.

**Definition 4.2** (Spectral Drinfeld class). Let  $A = B^{\text{ord}}(V^k(\mathfrak{g}))$  carry a spectral quasi-factorization datum with quasi-associator

$$\Phi(w, z) = 1 + \phi_p(w, z) \pmod{F^{p+1}},$$

where  $\phi_p$  is the first non-trivial term at filtration  $p$ . The spectral cochain complex is

$$\begin{aligned} C_{\text{spec}}^2(A) &= \{g(u) \in A^{\widehat{\otimes}^2}\}, \\ C_{\text{spec}}^3(A) &= \{\phi(w, z) \in A^{\widehat{\otimes}^3}\}, \end{aligned}$$

with linearized spectral differential

$$(\delta g)(w, z) = 1 \otimes g(z) + (\text{id} \otimes \Delta_z)g(w) - (\Delta_w \otimes \text{id})g(z) - g(w) \otimes 1. \quad (4.1)$$

The *spectral Drinfeld class at filtration  $p$*  is

$$\mathfrak{D}_{\text{spec}}(A, \Omega)|_{F^p} := [\phi_p] \in H_{\text{spec}}^3(\text{gr}^p A). \quad (4.2)$$

*Remark 4.3* (Interpretation). The spectral Drinfeld class measures the failure of the  $A_\infty$  bar-complex structure to strictify to a genuine associative product with spectral parameter. At each filtration  $p \geq 2$ , the pentagon relation on the quasi-associator linearizes:  $\phi_p$  is a 3-cocycle, a spectral twist shifts it by a coboundary, and the class  $[\phi_p] \in H_{\text{spec}}^3(\text{gr}^p A)$  is the intrinsic obstruction. If  $[\phi_p] = 0$  for all  $p \geq 2$ , an iterative filtered twist produces  $\Phi^G = 1$ , and the quasi-factorization datum strictifies to an honest spectral factorization  $R$ -matrix.

**4.4. Root-space one-dimensionality.** The key structural input is a classical theorem on the root structure of simple Lie algebras: every root space is one-dimensional. This forces the spectral Drinfeld class, which is valued in root spaces, to be a scalar, and a scalar is determined by a single equation.

**Theorem 4.4** (Root-space one-dimensionality). *Let  $\mathfrak{g}$  be a simple Lie algebra. For any subset  $S = \{\alpha_1, \dots, \alpha_n\}$  of simple roots such that  $\alpha = \sum_{i=1}^n \alpha_i$  is a root of  $\mathfrak{g}$ , the multilinear space*

$$\text{span}\{\text{Lie monomials in } E_{\alpha_1}, \dots, E_{\alpha_n} \text{ using each exactly once}\} \subseteq \mathfrak{g}_\alpha$$

*is one-dimensional.*

*Proof.* Every multilinear Lie monomial in  $E_{\alpha_1}, \dots, E_{\alpha_n}$  has weight  $\alpha = \alpha_1 + \dots + \alpha_n$ . In a simple Lie algebra, every root has multiplicity one:  $\dim \mathfrak{g}_\alpha = 1$ . The multilinear space is at most one-dimensional. At least one multilinear monomial is nonzero: since  $\alpha$  is a root and the  $\alpha_i$  are linearly independent, there exists an ordering  $\alpha_{i_1}, \dots, \alpha_{i_n}$  and a sequence of partial sums  $\beta_k = \alpha_{i_1} + \dots + \alpha_{i_k}$  such that each  $\beta_k$  is a root (by connectedness of the support in the Dynkin diagram), and the iterated bracket  $[E_{\alpha_{i_1}}, [E_{\alpha_{i_2}}, \dots, E_{\alpha_{i_n}}] \dots]$  is nonzero.  $\square$

#### 4.5. Vanishing of the obstruction.

**Theorem 4.5** (Spectral Drinfeld strictification). *For any simple Lie algebra  $\mathfrak{g}$ , on the evaluation-generated core  $\text{Core}_{\text{eval}}(\widehat{\mathfrak{g}}_k)$  the spectral Drinfeld class  $\mathfrak{D}_{\text{spec}}(A, \Omega)|_{F^p}$  vanishes at every filtration  $p \geq 2$ . The  $A_\infty$  structure on the ordered bar complex  $B^{\text{ord}}(V^k(\mathfrak{g}))$  strictifies to a genuine associative product with spectral parameter: there exists a spectral twist  $G$  such that the twisted quasi-associator  $\Phi^G = 1$ .*

*Remark 4.6* (Full-category extension). The eval-core restriction in Theorems 3.1, 3.3, 4.5, and 5.3 is : the bar-cobar identifications are proved on  $\text{Core}_{\text{eval}}(\widehat{\mathfrak{g}}_k)$ , where the MC3 cocycle conditions are unconditional. Extension to the full category of smooth  $V^k(\mathfrak{g})$ -modules (beyond evaluation modules and their tensor generates) is conjectural; the downstream DK-4 and DK-5 identifications are correspondingly conditional on this extension.

*Proof.* The proof is an induction on filtration.

*Base case* ( $p = 2$ ). The spectral Drinfeld class at  $F^2$  is valued in  $H_{\text{spec}}^3(\text{gr}^2 A)$ . The associated graded  $\text{gr}^2 A$  decomposes by root sectors. In each root sector  $\mathfrak{g}_\alpha$  (with  $\alpha = \alpha_i + \alpha_j$  a sum of two simple roots), the target space is  $\mathfrak{g}_\alpha \otimes \mathfrak{g}_\beta \otimes \mathfrak{g}_\gamma$  for appropriate roots. By root-space one-dimensionality (Theorem 4.4), each factor is one-dimensional, so the class is a scalar. The co-product rigidity (the spectral differential (4.1) has a unique scalar coboundary at each root sector) forces this scalar to vanish.

*Inductive step* ( $p \rightarrow p+1$ ). Assume  $\Phi|_{F^p} = 1$  after a spectral twist. The class  $[\phi_{p+1}] \in H_{\text{spec}}^3(\text{gr}^{p+1} A)$  is again valued in root spaces. The multilinear Lie monomials of weight  $\alpha_1 + \dots + \alpha_{p+1}$  span a one-dimensional space in

$\mathfrak{g}_\alpha$  by Theorem 4.4. The Jacobi identity collapses any multi-dimensional contributions from the free Lie algebra to this one-dimensional target.

The key computation: for the  $D_4$  star (four simple roots  $\alpha_1, \alpha_2, \alpha_3, \alpha_c$  with  $\alpha_c$  adjacent to each  $\alpha_i$  and the  $\alpha_i$  mutually non-adjacent), the free Lie algebra modulo non-adjacency commutation has a two-dimensional multilinear space at degree 4. In  $\mathfrak{g}$ , the Jacobi identity  $[X_2, [X_c, X_3]] = [X_3, [X_c, X_2]]$  (from  $[X_2, X_3] = 0$ ) collapses the two generators  $A = [[X_2, [X_1, X_c]], X_3]$  and  $B = [[X_3, [X_1, X_c]], X_2]$  to  $A = B$ . The coproduct rigidity then forces the scalar obstruction to vanish, as at  $p = 2$ .

The induction produces a sequence of spectral twists  $G_2, G_3, \dots$  whose composition converges in the completed filtered algebra and gives the desired  $G$  with  $\Phi^G = 1$ .  $\square$

*Remark 4.7* (The obstruction has nowhere to live). The vanishing of the spectral Drinfeld class is not a lucky cancellation. The obstruction is valued in root spaces, and root spaces are one-dimensional for simple Lie algebras. A cohomology class valued in a one-dimensional space is a scalar. A scalar is determined by a single equation. The coproduct rigidity provides that equation. The obstruction vanishes because the target is too small to support a non-trivial class.

This mechanism fails for Kac–Moody algebras  $\mathfrak{g}$  that are not finite-dimensional simple: when root multiplicities exceed one, the target space can be higher-dimensional, the Jacobi collapse fails, and the strictification problem acquires genuine obstructions.

**4.6. From  $A_\infty$  to the Yangian.** The strictified product with spectral parameter is the Yangian multiplication.

**Corollary 4.8** (Yangian as strictified bar product). *The strictified ordered bar complex of  $V^k(\mathfrak{g})$  is the Yangian  $Y_\hbar(\mathfrak{g})$  at  $\hbar = \pi i/(k + h^\vee)$ : the spectral twist  $G$  of Theorem 4.5 converts the  $A_\infty$  bar-complex product into the Yangian product  $m_\hbar(u)$ , and the monodromy of the strictified connection is the Yangian  $R$ -matrix  $R^Y(u)$ .*

*Proof.* The strictified product satisfies the spectral associativity  $(m_\hbar \otimes \text{id}) \circ m_\hbar = (\text{id} \otimes m_\hbar) \circ m_\hbar$  and the RTT relation  $R_{12}(u - v) T_1(u) T_2(v) = T_2(v) T_1(u) R_{12}(u - v)$ , where  $T(u) = \sum_{k \geq 0} T_k u^{-k}$  is the generating matrix. These are the defining relations of the Yangian  $Y_\hbar(\mathfrak{g})$ . The identification with the standard Yangian (generated by  $\mathfrak{g}[t]$  with the Drinfeld coproduct) follows from the Drinfeld new realization theorem.  $\square$

## 5. DK-2: THE DG-SHIFTED YANGIAN AS KOSZUL DUAL

**5.1. The dg-shifted Yangian.** The spectral Drinfeld strictification produces a strict associative algebra with spectral parameter from the bar complex. This algebra deserves its own name.

**Definition 5.1** (The dg-shifted Yangian). The *dg-shifted Yangian*  $Y_h^{\text{dg}}(\mathfrak{g})$  is the dg algebra whose underlying graded algebra is the Yangian  $Y_h(\mathfrak{g})$  and whose differential  $d$  is the residual bar differential after strictification: the component of  $d_B$  that survives the spectral twist  $G$  of Theorem 4.5.

On the associated graded with respect to the PBW filtration,  $d$  acts as the Chevalley–Eilenberg differential of the current Lie algebra  $\mathfrak{g}[t]$ :  $\text{gr } Y_h^{\text{dg}}(\mathfrak{g}) \simeq (\text{CE}(\mathfrak{g}[t]), d_{\text{CE}})$ .

*Remark 5.2* (Four Yangian types). The dg-shifted Yangian  $Y_h^{\text{dg}}(\mathfrak{g})$  is distinct from the classical Yangian  $Y_h(\mathfrak{g})$ , the chiral Yangian  $Y(\mathfrak{g})^{\text{ch}}$ , and the spectral Yangian on  $\mathbb{A}_u^1$ .

- (i) The *classical Yangian*  $Y_h(\mathfrak{g})$  is an  $\mathbf{E}_1$ -topological algebra on  $\mathbb{R}$ .
- (ii) The *dg-shifted Yangian*  $Y_h^{\text{dg}}(\mathfrak{g})$  lives at a point or on the formal disk, from Koszul duality.
- (iii) The *chiral Yangian*  $Y(\mathfrak{g})^{\text{ch}}$  is an  $\mathbf{E}_1$ -chiral algebra on a curve  $X$ , a  $\mathcal{D}$ -module with  $R$ -twisted OPE.
- (iv) The *spectral Yangian* is a factorization algebra on  $\mathbb{A}_u^1$ .

The passage from (ii) to (iii) is the globalization problem; the passage from (i) to (iv) is the spectral parametrization. Conflating any two is a type error.

**5.2. Koszul duality identification.** The dg-shifted Yangian is the  $\mathbf{E}_1$ -chiral Koszul dual of the affine algebra.

**Theorem 5.3** (Yangian as Koszul dual). *For  $\mathfrak{g}$  simple and  $k \neq -h^\vee$  generic, on the evaluation-generated core  $\text{Core}_{\text{eval}}(\widehat{\mathfrak{g}}_k)$  the  $\mathbf{E}_1$ -chiral Koszul duality functor applied to the affine vertex algebra  $V^k(\mathfrak{g})$  produces the dg-shifted Yangian:*

$$(V^k(\mathfrak{g}))^{!,\mathbf{E}_1} \simeq Y_h^{\text{dg}}(\mathfrak{g}), \quad \hbar = \frac{\pi i}{k + h^\vee}. \quad (5.1)$$

The Koszul duality interchanges:

| <i>Affine side</i> $(V^k(\mathfrak{g}))$         | <i>Yangian side</i> $(Y_h^{\text{dg}}(\mathfrak{g}))$  |
|--------------------------------------------------|--------------------------------------------------------|
| <i>Chiral OPE</i> ( $\mathbf{E}_\infty$ -chiral) | <i>Spectral product</i> ( $\mathbf{E}_1$ -topological) |
| <i>Level</i> $k$                                 | <i>Planck constant</i> $\hbar = \pi i / (k + h^\vee)$  |
| <i>KZ monodromy</i>                              | <i>R-matrix</i>                                        |
| <i>Affine Weyl group</i>                         | <i>Quantum Weyl group</i>                              |
| <i>Critical level</i> $k = -h^\vee$              | <i>Classical limit</i> $\hbar = \infty$                |

*Proof.* The  $\mathbf{E}_1$ -chiral Koszul dual of  $\mathcal{A}$  is defined as the cohomology of the ordered bar complex:  $(V^k(\mathfrak{g}))^{!,\mathbf{E}_1} = H^*(B^{\text{ord}}(V^k(\mathfrak{g})))$ , endowed with the algebra structure from the deconcatenation coproduct and the spectral product from Theorem 4.5.

The identification with  $Y_h^{\text{dg}}(\mathfrak{g})$  proceeds in three steps.

*Step 1: PBW.* The PBW theorem for the Yangian gives a filtration on  $Y_h(\mathfrak{g})$  whose associated graded is  $U(\mathfrak{g}[t])$ . The corresponding filtration on

$B^{\text{ord}}(V^k(\mathfrak{g}))$  is the collision-depth filtration, and the associated graded of the bar cohomology is the Chevalley–Eilenberg cohomology  $H^*(\mathfrak{g}[t], \mathbb{C})$ . By the Whitehead reduction (for the finite-dimensional simple Lie algebra  $\mathfrak{g}$ , not for  $\mathfrak{g}[t]$  directly; the reduction uses the semisimplicity of  $\mathfrak{g}$ ), this cohomology is concentrated in the expected degrees.

*Step 2: Product identification.* The strictified product (Corollary 4.8) on bar cohomology satisfies the Yangian defining relations (RTT form) by construction. The spectral parameter  $u$  enters via the collision residue: the insertion points  $z_1, \dots, z_n$  of the bar complex become the spectral parameters of the Yangian generators after the  $d \log$  kernel absorption.

*Step 3: Differential.* The residual differential on  $Y_h^{\text{dg}}(\mathfrak{g})$  is the component of  $d_B$  that does not contribute to the product structure. On the associated graded, this is the Chevalley–Eilenberg differential  $d_{\text{CE}}$  of  $\mathfrak{g}[t]$ . The full differential includes higher-order corrections controlled by the higher collision residues.  $\square$

*Remark 5.4 (Line-side interpretation).* In the holographic dictionary of the  $E_1$ -chiral programme, the affine algebra  $V^k(\mathfrak{g})$  lives on the “bulk” side (the chiral algebra on the curve), while the Yangian  $Y_h^{\text{dg}}(\mathfrak{g})$  lives on the “line” side (the line operator algebra on  $\mathbb{R}$ ). The Koszul duality  $(V^k(\mathfrak{g}))^{!, E_1} = Y_h^{\text{dg}}(\mathfrak{g})$  is the algebraic incarnation of the holographic correspondence: bulk chiral algebras are Koszul dual to boundary quantum groups.

## 6. DK-3: COMPLETE STRICTIFICATION FOR ALL SIMPLE TYPES

**6.1. The type dependence.** The strictification of Theorem 4.5 is proved uniformly for all simple Lie algebras. The mechanism (root-space one-dimensionality plus Jacobi collapse) works identically across all Dynkin types  $A_n, B_n, C_n, D_n$ , and the exceptionals  $G_2, F_4, E_6, E_7, E_8$ .

The type enters only through the combinatorics of the root system: which subsets of simple roots sum to a root, which adjacency patterns appear in the Dynkin diagram, and how the Jacobi identity collapses the free Lie algebra to  $\mathfrak{g}$ .

## 6.2. Type-by-type verification.

**Proposition 6.1** (Strictification by Dynkin type). *For each simple Lie algebra  $\mathfrak{g}$ , the spectral Drinfeld class vanishes at all filtration levels. The type-specific content is:*



| <i>Type</i>     | <i>Rank</i> | <i>Max. branching</i>          | <i>Collapse mechanism</i>                      |
|-----------------|-------------|--------------------------------|------------------------------------------------|
| $A_n$           | $\geq 1$    | <i>path (no branching)</i>     | <i>Path one-dimensionality</i>                 |
| $B_n, C_n$      | $\geq 2$    | <i>single short/long root</i>  | <i>Path + Jacobi at double bond</i>            |
| $D_n$           | $\geq 4$    | <i>trivalent vertex</i>        | <i>Jacobi collapse (Lemma below)</i>           |
| $G_2$           | 2           | <i>triple bond</i>             | <i>Direct computation</i>                      |
| $F_4$           | 4           | <i>double bond + branching</i> | <i><math>D_4</math>-star + Jacobi</i>          |
| $E_6, E_7, E_8$ | 6, 7, 8     | <i>trivalent vertex</i>        | <i><math>D_4</math>-star + Jacobi collapse</i> |

In types  $A_n$ ,  $B_n$ ,  $C_n$ , the Dynkin diagram is a path and the multilinear Lie space is one-dimensional without needing Jacobi. In types  $D_n$  and  $E_n$ , the trivalent vertex of the Dynkin diagram creates a two-dimensional free Lie space that the Jacobi identity collapses to one. Type  $G_2$  is verified by direct computation (the root system has 12 roots and the strictification involves finitely many sectors).

*Proof.* For type  $A_n$ : the Dynkin diagram is a path  $\alpha_1 - \alpha_2 - \dots - \alpha_n$ . Any root  $\alpha_i + \alpha_{i+1} + \dots + \alpha_j$  is a sum of consecutive simple roots. The multilinear Lie monomial  $[E_{\alpha_i}, [E_{\alpha_{i+1}}, \dots, E_{\alpha_j}] \dots]$  is the unique nonzero element (up to scalar) by the path one-dimensionality of iterated brackets in a simple Lie algebra with a path-shaped support. No Jacobi collapse is needed.

For type  $D_n$  ( $n \geq 4$ ): the trivalent vertex  $\alpha_{n-2}$  is adjacent to  $\alpha_{n-3}$ ,  $\alpha_{n-1}$ , and  $\alpha_n$ . The  $D_4$ -star configuration (with  $\alpha_c = \alpha_{n-2}$  and  $\alpha_1, \alpha_2, \alpha_3 = \alpha_{n-3}, \alpha_{n-1}, \alpha_n$ ) produces a two-dimensional space in the free Lie algebra. The Jacobi identity:  $[X_2, [X_c, X_3]] = [X_3, [X_c, X_2]]$  (from  $[X_2, X_3] = 0$ ) collapses the two generators  $A = [[X_2, [X_1, X_c]], X_3]$  and  $B = [[X_3, [X_1, X_c]], X_2]$  via the chain  $A = [X_1, [X_2, [X_c, X_3]]] = [X_1, [X_3, [X_c, X_2]]] = B$ . The multilinear space is one-dimensional in  $\mathfrak{g}$ , the obstruction is a scalar, and coproduct rigidity forces it to vanish.

For types  $E_6, E_7, E_8$ : the Dynkin diagram has a single trivalent vertex, and the analysis reduces to the  $D_4$ -star case at that vertex. All other sectors are path-shaped and the multilinear space is one-dimensional without Jacobi.

Type  $G_2$  has two simple roots  $\alpha_1, \alpha_2$  connected by a triple bond. The root system has six positive roots:  $\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2, 3\alpha_1 + \alpha_2, 3\alpha_1 + 2\alpha_2$ . At each multilinear sector, root-space one-dimensionality gives a one-dimensional target, the obstruction is scalar, and it vanishes by coproduct rigidity.

Type  $F_4$  combines a double bond and a trivalent structure (the Dynkin diagram  $\alpha_1 - \alpha_2 \Rightarrow \alpha_3 - \alpha_4$  has no branching, but the double bond at  $\alpha_2 - \alpha_3$  creates short-root sectors with multiplicity patterns analogous to the  $B_n$  case). The analysis follows from the  $B_n$  and  $D_4$ -star mechanisms.  $\square$

**6.3. Extension to the evaluation-generated core.** The strictification extends to the full evaluation-generated subcategory of modules.

**Corollary 6.2** (DK-3 on the evaluation-generated core). *The  $E_1$ -chiral Koszul duality functor restricts to an equivalence of  $E_1$ -factorization categories on the evaluation-generated cores:*

$$\Phi: \text{Fact}_{E_1}^{\text{eval}}(Y_h(\mathfrak{g})) \xrightarrow{\sim} \text{Fact}_{E_1}^{\text{eval}}(Y_{-h}(\mathfrak{g}))^{\text{op}}$$

for all simple Lie algebras  $\mathfrak{g}$ .

*Proof.* The type-uniform strictification (Proposition 6.1) provides the  $A_\infty$ -to-strict comparison at each filtration level. On evaluation modules  $V(a_1) \otimes \cdots \otimes V(a_n)$ , the higher  $A_\infty$  operations do not contribute (one-loop collapse), so the comparison reduces to the DK-0 monodromy identification (Theorem 3.1). The factorization structure is preserved by the Verdier involution on ordered configuration spaces (Theorem 3.3).  $\square$

**Conjecture 6.3** (Full DK-3 beyond evaluation modules). *The equivalence of Corollary 6.2 extends to the full  $E_1$ -factorization categories:*

$$\text{Fact}_{E_1}(\mathcal{Y}(\mathfrak{g})) \simeq \text{Fact}_{E_1}(U_q(\mathfrak{g}))^{\text{op}},$$

where  $\mathcal{Y}(\mathfrak{g})$  is the Yangian  $E_1$ -chiral algebra. The extension beyond evaluation objects requires the thick generation argument: the evaluation-generated subcategory must be shown to thickly generate the full category. In type  $A$ , this reduces to explicit Clebsch–Gordan computations; for other types, it is open.

## 7. THE HEISENBERG ALGEBRA: THE TRIVIAL BRIDGE

The Heisenberg algebra  $\mathcal{H}_k$  is the simplest benchmark: the Lie bracket vanishes, the collision residue is scalar, and the Drinfeld–Kohno bridge is trivial.

**7.1. The collision residue.** The OPE of the Heisenberg current is  $J(z)J(w) \sim k/(z-w)^2$ . The double pole, after  $d \log$  absorption, yields the collision residue

$$r^{\text{Heis}}(z) = \frac{k}{z}. \quad (7.1)$$

This is scalar: there is no Casimir tensor because the Lie bracket vanishes.

**7.2. The Kohno connection.** The Kohno connection on  $\text{Conf}_n^{\text{ord}}(\mathbb{C})$  with fibre  $\mathbb{C}$  (the trivial  $\mathcal{H}_k$ -module) is

$$\nabla_{\text{Heis}} = d - \sum_{i < j} \frac{k}{z_i - z_j} d(z_i - z_j).$$

This is an abelian connection (the fibre is one-dimensional), so the monodromy is the scalar exponential.

**7.3. The  $R$ -matrix.** The monodromy around the braid generator  $\gamma_{12}$  ( $z_1$  circles  $z_2$ ) is

$$R^{\text{Heis}} = \exp\left(\oint_{\gamma} \frac{k}{z} dz\right) = e^{2\pi i k}.$$

This is a scalar phase: the quantum group  $U_q(\text{Heis})$  is commutative, the  $R$ -matrix is  $e^{k\hbar/z}$  (as a formal power series in  $1/z$ ), and the braid representation is abelian.

**7.4. The shadow invariant.** The modular characteristic of the Heisenberg algebra is  $\kappa(\mathcal{H}_k) = k$ . The averaging map  $\text{av}(r(z)) = \text{av}(k/z) = k = \kappa$  is exact: for abelian algebras,  $\text{av}(r) = \kappa$  without Sugawara correction. The Drinfeld-Kohno bridge is trivial because all the non-trivial content (CYBE, associator, quantum group) requires non-abelian structure.

## 8. THE $\mathfrak{sl}_2$ EXAMPLE: THE NON-TRIVIAL BRIDGE

For  $\mathfrak{g} = \mathfrak{sl}_2$ , the Drinfeld-Kohno bridge is non-trivial: the collision residue is matrix-valued, the Kohno connection is non-abelian, and the  $R$ -matrix is a  $4 \times 4$  matrix on the tensor square of the fundamental representation.

**8.1. Setup.** Let  $\mathfrak{g} = \mathfrak{sl}_2$  with standard basis  $\{E, F, H\}$  satisfying  $[H, E] = 2E$ ,  $[H, F] = -2F$ ,  $[E, F] = H$ . The Killing form is  $\langle X, Y \rangle = 4 \text{tr}(XY)$  in the fundamental representation, and the Casimir tensor in the orthonormal basis  $\{I^a\}$  is

$$\Omega = \frac{1}{2}(E \otimes F + F \otimes E + \frac{1}{2}H \otimes H).$$

The dual Coxeter number is  $h^\vee = 2$ , so the KZ parameter is  $\hbar = \pi i/(k+2)$  and  $q = e^{\pi i/(k+2)}$ .

**8.2. The collision residue and KZ connection.** The collision residue for  $\widehat{\mathfrak{sl}}_2$  at level  $k$  is

$$r(z) = \frac{k \Omega}{z}$$

in the trace form, or equivalently  $\Omega/((k+2)z)$  in the KZ normalization. The KZ connection on  $\text{Conf}_n(\mathbb{C})$  with fibre  $V_{j_1} \otimes \cdots \otimes V_{j_n}$  is

$$\nabla_{\text{KZ}} = d - \frac{1}{k+2} \sum_{i < j} \Omega_{ij} \frac{dz_{ij}}{z_{ij}}.$$

**8.3. The  $4 \times 4$   $R$ -matrix.** On the tensor square  $V_{1/2} \otimes V_{1/2}$  of the fundamental representation, the Casimir  $\Omega$  acts with eigenvalue  $+1/4$  on the symmetric (triplet,  $j = 1$ ) component and  $-3/4$  on the antisymmetric (singlet,  $j = 0$ ) component. The KZ monodromy is

$$R = \exp\left(\frac{2\pi i}{k+2} \Omega\right) = q^\Omega, \quad q = \exp\left(\frac{2\pi i}{k+2}\right), \quad (8.1)$$

so that on eigencomponents

$$R = \begin{pmatrix} q^{1/4} & 0 & 0 & 0 \\ 0 & q^{1/4} \cos \theta & i q^{1/4} \sin \theta & 0 \\ 0 & i q^{1/4} \sin \theta & q^{1/4} \cos \theta & 0 \\ 0 & 0 & 0 & q^{1/4} \end{pmatrix} \quad (8.2)$$

in the basis  $\{v_+ \otimes v_+, v_+ \otimes v_-, v_- \otimes v_+, v_- \otimes v_-\}$ , where  $\theta$  depends on the normalization of the contour integral.

More precisely, in the singlet-triplet basis, the  $R$ -matrix diagonalizes to

$$R = \text{diag}(\underbrace{q_D^{1/2}, q_D^{1/2}, q_D^{1/2}}_{\text{triplet}}, \underbrace{-q_D^{-3/2}}_{\text{singlet}}), \quad (8.3)$$

written in the Drinfeld normalization  $q_D = e^{i\pi/(k+2)}$  so that  $q_D^2 = q$  equals the Kazhdan–Lusztig root of unity used in equations (8.1)–(8.2). The relative phase between the singlet and triplet is  $-q_D^{-2} = -q^{-1}$ , which is the fundamental building block of the colored Jones polynomial.

**8.4. Four-point conformal blocks.** For  $n = 4$  with all fundamental external legs, the KZ equation is the hypergeometric ODE (3.3). The two solutions  $F_0$  (singlet channel) and  $F_1$  (triplet channel) have monodromy  $\text{diag}(1, e^{4\pi i/(k+2)})$  around  $z = 0$ , and the braiding matrix in the  $s$ -channel basis is

$$B_s = \begin{pmatrix} -q^{-3/2} & 0 \\ 0 & q^{1/2} \end{pmatrix},$$

confirming the Drinfeld–Kohno identification at the eigenvalue level.

**8.5. Colored Jones from bar monodromy.** The colored Jones polynomial  $J_n(K; q)$  of a knot  $K$  colored by the  $n$ -dimensional representation  $V_{(n-1)/2}$  of  $\mathfrak{sl}_2$  is computed by the closure of the braid representation.

**Proposition 8.1** (Colored Jones from bar-complex monodromy). *Let  $K$  be a knot presented as the closure of a braid  $\beta \in B_m$ . The colored Jones polynomial is*

$$J_n(K; q) = \frac{\text{tr}_{V_{(n-1)/2}^{\otimes m}}(\rho_m^{\text{KZ}}(\beta))}{\text{tr}_{V_{(n-1)/2}^{\otimes m}}(\text{id})}, \quad (8.4)$$

where  $\rho_m^{\text{KZ}}$  is the KZ monodromy representation at level  $k$  with  $q = e^{i\pi/(k+2)}$ , and the trace requires the Markov normalization (writhe correction and quantum dimension factor).

*Proof.* The KZ monodromy representation  $\rho_m^{\text{KZ}}$  on  $V_{(n-1)/2}^{\otimes m}$  coincides with the quantum group braid representation  $\rho_m^{U_q}$  by the Drinfeld–Kohno theorem (Theorem 3.1). The colored Jones polynomial is the quantum group invariant of  $K$ : the trace of the braid representation with Markov normalization. The raw trace of the braid closure gives the unnormalized invariant; the Markov normalization (writhe correction via  $q^{\pm w(K) \cdot C_2}$  and division by the quantum

dimension  $[n]_q = (q^{n/2} - q^{-n/2}) / (q^{1/2} - q^{-1/2})$  produces the standard colored Jones polynomial.

In the bar-complex language: the braid closure corresponds to the identification of insertion points on a closed curve (genus  $\geq 1$ ), and the trace is the genus-1 partition function of the ordered bar complex. The writhe correction is the framing anomaly from the quadratic Casimir.  $\square$

*Remark 8.2* (The non-trivial content). For  $\mathfrak{sl}_2$ , the entire Drinfeld–Kohno bridge is explicit: the KZ equation is hypergeometric, the  $R$ -matrix eigenvalues are  $q^{1/2}$  and  $-q^{-3/2}$ , the associator has an explicit formula as a formal power series in  $\Omega_{12}$  and  $\Omega_{23}$ , and the colored Jones polynomial is computable from the monodromy matrices.

The bar-complex perspective adds the following: the collision residue  $r(z) = k\Omega/z$  is the degree-2 projection of the Maurer–Cartan element  $\Theta_{\mathcal{A}} \in \mathfrak{g}_{\mathcal{A}}^{\mathbf{E}_1}$ ; the CYBE is  $d_B^2 = 0$  at degree 2; the associator is the degree-3 shadow; and the colored Jones polynomial is the genus-1 trace of the  $\mathbf{E}_1$  monodromy. The bar complex holds all these simultaneously, and the Drinfeld–Kohno theorem is the statement that the degree-2 projection suffices to reconstruct the quantum group on evaluation modules.

## REFERENCES

- [1] V. G. Drinfeld, *Hopf algebras and the quantum Yang–Baxter equation*, Soviet Math. Dokl. **32** (1985), 254–258.
- [2] V. G. Drinfeld, *On quasitriangular quasi-Hopf algebras and on a group that is closely connected with  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$* , Leningrad Math. J. **2** (1991), 829–860.
- [3] P. Etingof, I. Frenkel, and A. Kirillov, *Lectures on representation theory and Knizhnik–Zamolodchikov equations*, Mathematical Surveys and Monographs **58**, American Mathematical Society, 1998.
- [4] T. Kohno, *Monodromy representations of braid groups and Yang–Baxter equations*, Ann. Inst. Fourier **37** (1987), 139–160.
- [5] D. Kazhdan and G. Lusztig, *Tensor structures arising from affine Lie algebras, I–IV*, J. Amer. Math. Soc. **6–7** (1993–1994).
- [6] V. G. Knizhnik and A. B. Zamolodchikov, *Current algebra and Wess–Zumino model in two dimensions*, Nuclear Phys. B **247** (1984), 83–103.

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